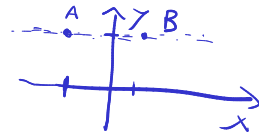
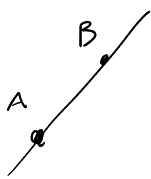


Seção 1.3 $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{6 - 18}{-4 - (-1)} = \frac{-12}{-3} = 4$



- 1 $A(-4, 6), B(-1, 18)$ 2 $A(6, -2), B(-3, 5)$ 3 $A(-1, -3), B(-1, 2)$ 4 $A(-3, 4), B(2, 4)$

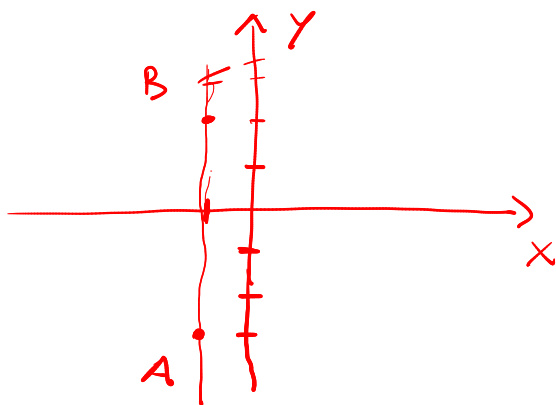


$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$1. m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{18 - 6}{-1 - (-4)} = \frac{12}{3} = 4$$

$$2) m = \frac{\Delta y}{\Delta x} = \frac{5 - (-2)}{-3 - 6} = \frac{7}{-9} = -\frac{7}{9}$$

$$3) m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2 - (-3)}{-1 - (-1)} = \frac{2 + 3}{0} (!?)$$

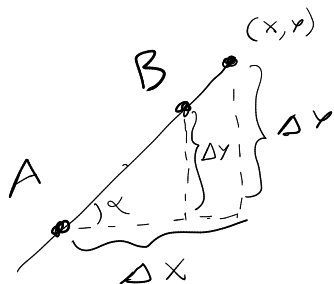


~~A~~ m, pois $\overline{AB} \perp \overrightarrow{OX}$

$$4) m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - 4}{2 - (-3)} = \frac{0}{5} = 0 \parallel \boxed{m = 0}$$

14 Intercepto-x - 4, intercepto-y 8.

$A = (-4, 0) \in R_{ETA}$ $B = (0, 8) \in R_{ETA}$



$$\frac{y - y_1}{x - x_1} = m \Leftrightarrow \boxed{y - y_1 = m(x - x_1)}$$

$$y - y_1 = 2(x - x_1)$$

$$y - 0 = 2(x + 4)$$

$$m = \frac{\Delta y}{\Delta x} = \frac{8 - 0}{0 - (-4)} = \frac{8}{4} = 2$$

$$\boxed{y = 2x + 8}$$

SE

$$(x_1, y_1) = B$$

$$y - y_1 = 2(x - x_1)$$

$$y - 8 = 2(x - 0)$$

$$y - 8 = 2x$$

$$y = 2x + 8$$

15 Por A (8, -2), intercepto-y -3.

$$\Rightarrow \boxed{b = -3}$$

$$y = \underbrace{a}_{\text{coef. angular}} x + \underbrace{b}_{\text{COEF. LINEAR}}$$

$$\hookrightarrow y = ax - 3$$

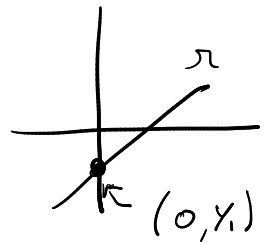
$$(8, -2) \in \text{RETA}$$

$$-2 = a \cdot 8 - 3$$

$$-2 + 3 = a \cdot 8$$

$$1 = 8 \cdot a \Rightarrow \boxed{a = \frac{1}{8}}$$

Logo, $\boxed{y = \frac{1}{8}x - 3}$



$$(0, y_1) \in r$$

$$y = ax + b$$

$$y_1 = a \cdot 0 + b$$

$$\boxed{y_1 = b}$$

25 $3x - 4y + 8 = 0$

$$\hookrightarrow y = ax + b$$

$$3x + 8 = +4y$$

$$\underline{4}y = 3x + 8$$

$$y = \frac{3x + 8}{4} = \frac{3x}{4} + \frac{8}{4}$$

$$\boxed{y = \frac{3}{4}x + 2}$$

cuta y
↓

37 Se uma reta l tem interceptos-y e x iguais a a e b , respectivamente (diferentes de zero), mostre que a equação de l pode escrever-se sob a forma

$$\frac{x}{a} + \frac{y}{b} = 1.$$

(É a forma segmentária da equação da reta.) Escreva nessa forma a equação $4x - 2y = 6$.

cuta y em $a \neq 0$

cuta x em $b \neq 0$

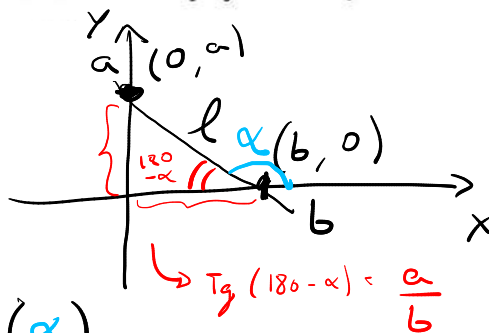
$$(0, a) \in l$$

$$(b, 0) \in l$$

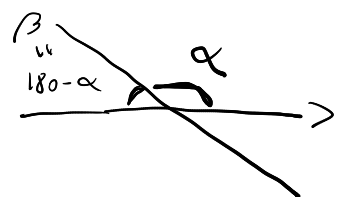
Por (*)

$$m_l = \text{Tg}(\alpha)$$

$$m_l = -\text{Tg}(180 - \alpha) = -\frac{a}{b}$$



LEMBRETE:



$$m = \text{Tg} \alpha$$

$$-\text{Tg}(180 - \alpha) = \text{Tg} \alpha$$

$$\text{Tg}(180 - \alpha) = -\text{Tg}(\alpha)$$

$$\underline{m}_l = -\frac{a}{b} \left\{ \begin{array}{l} \text{corta } y \\ \text{corta } x \end{array} \right\} \Rightarrow y = \underline{m}_l x + a$$

$$\boxed{y = -\frac{a}{b}x + a}$$

$$\frac{y}{b} = \left(-\frac{ax}{b} + a\right) \cdot \frac{1}{b} \Rightarrow \frac{y}{b} = -\frac{ax}{b^2} + \frac{a}{b}$$

$$\frac{b}{a} \times \left(\frac{ax}{b^2} + \frac{y}{b} \right) = \left(\frac{a}{b} \right) \times \frac{b}{a}$$

$$\boxed{\frac{x}{b} + \frac{y}{a} = 1}$$

$$4x - 2y = 6$$

$$y = 0 \Rightarrow 4x = 6 \Rightarrow x = \frac{3}{2} \leftarrow \text{Interceptor } x \Rightarrow b$$

$$x = 0 \Rightarrow -2y = 6 \Rightarrow y = -3 \leftarrow \text{Interceptor } y \Rightarrow a$$

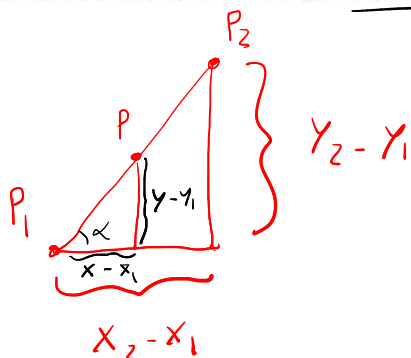
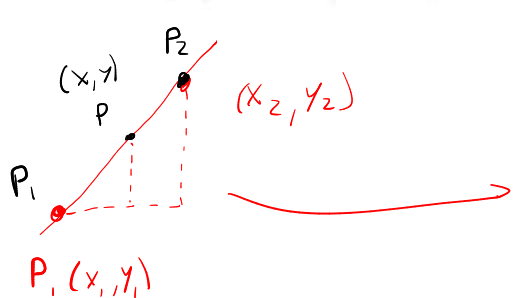
$$4x - 2y = 6 \Leftrightarrow \frac{4x - 2y}{6} = \frac{6}{6} \Leftrightarrow \frac{4x}{6} - \frac{2y}{6} = 1$$

$$\Leftrightarrow \frac{x}{\frac{6}{4}} + \frac{y}{-\frac{6}{2}} = 1$$

38 Prove que a equação da reta por $P_1(x_1, y_1)$ e $P_2(x_2, y_2)$ é

$$(y - y_1)(x_2 - x_1) = (y_2 - y_1)(x - x_1).$$

(É a equação da reta por dois pontos.) Use-a para escrever a equação da reta por $A(7, -1)$ e $B(4, 6)$.



$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$(y - y_1)(x_2 - x_1) = (y_2 - y_1)(x - x_1)$$

$$A = \begin{pmatrix} x_1 & y_1 \\ 7 & -1 \end{pmatrix} \Rightarrow (y - (-1))(4 - 7) = (6 - (-1))(x - 7)$$

$$B = \begin{pmatrix} x_2 & y_2 \\ 4 & 6 \end{pmatrix}$$

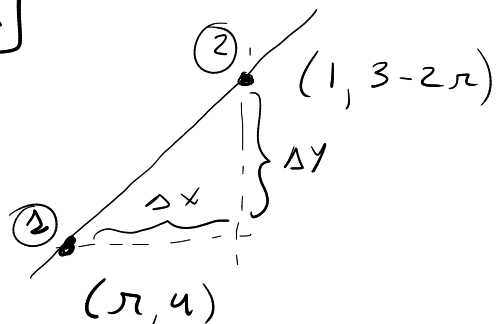
$$(y + 1)(-3) = 7(x - 7)$$

Obs: $y + 1 = \left(-\frac{7}{3}\right)(x - 7)$

39 Determine todos os valores de r , tais que o coeficiente angular da reta pelos pontos $(r, 4)$ e $(1, 3 - 2r)$ seja menor do que 5.

40 Determine todos os valores de t , tais que o coeficiente angular da reta por $(t, 3t + 1)$ e $(1 - 2t, t)$ seja maior do que 4.

39]



$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

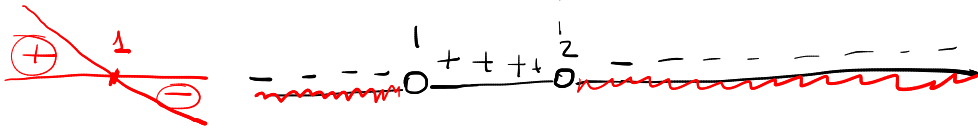
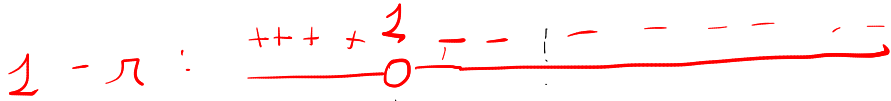
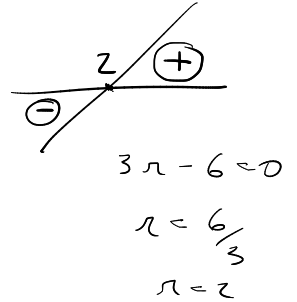
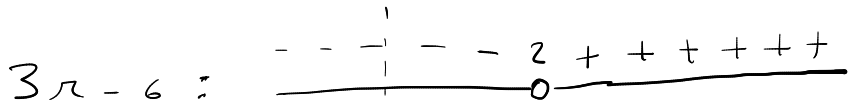
$$= \frac{(3 - 2r) - 4}{1 - r} = \frac{-2r - 1}{1 - r}$$

Queremos $m < 5$, ou seja:

$$\frac{-2r - 1}{1 - r} < 5 \quad \frac{-2r - 1}{1 - r} - 5 < 0$$

$$\frac{-2\pi - 1}{1 - \pi} - 5 \times \frac{(1 - \pi)}{(1 - \pi)} < 0$$

$$\frac{-2\pi - 1 - 5 + 6\pi}{1 - \pi} < 0 \Leftrightarrow \frac{3\pi - 6}{1 - \pi} < 0$$



$$\frac{3\pi - 6}{1 - \pi}$$

$$S = (-\infty, 1) \cup (2, +\infty)$$

40]

$$(x_2, y_2) = (t, 3t+1)$$

$$(1-2t, t)$$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{(3t+1) - t}{t - (1-2t)}$$

$$= \frac{2t + 1}{3t - 2} = m$$

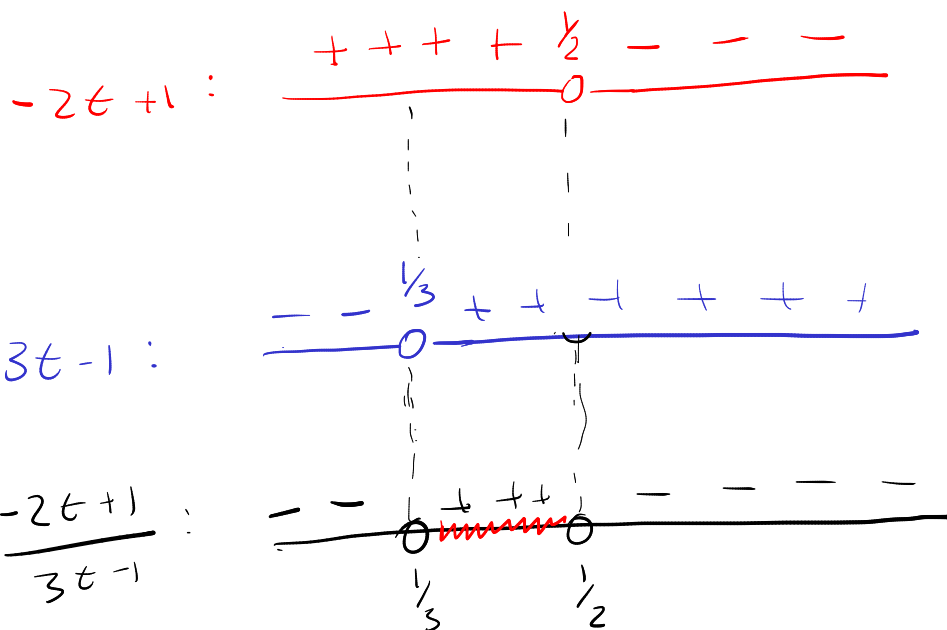
Queremos $m > 4$

$$\frac{2t+1}{3t-2} > 4 \Leftrightarrow \frac{2t+1}{3t-1} - 4 > 0$$

$$\frac{2t+1}{3t-1} - \frac{4(3t-1)}{3t-1} > 0$$

$$\frac{2t+1-12t+4}{3t-1} > 0 \quad \frac{1}{5} \times \frac{-10t+5}{3t-1} > 0 \times \frac{1}{5}$$

$$\frac{-2t+1}{3t-1} > 0$$



$$S = \left(\frac{1}{3}, \frac{1}{2}\right)$$

